LECTURE 10

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$$\sum \frac{1}{n^2}$$

$$\underbrace{\sum \frac{1}{(n+1)^2}}_{\text{finite}} < \sum \frac{1}{n^2} \to \text{finite}$$

$$\underbrace{\sum \frac{1}{n-10}}_{\text{infinite}} > \sum \frac{1}{n} \to \text{sum} = \infty$$

div.

$$\sum \frac{n}{n^2+1} \sim \sum \frac{1}{n} = b_n \qquad b_n = 1/n$$

$$\lim \frac{a_n}{b_n} = \frac{n^2}{n^2+1} \to 1 \neq 0 \text{ \& finite}$$

$$\sum \frac{1}{\sqrt{5n^2+7n+11}} \sim \sum \left(\frac{1}{n}\right) = b_n$$

$$\lim \frac{a_n}{b_n} = \frac{1}{\sqrt{5}} \neq 0 \text{ finite}$$

$$\sum_{\text{cgt}} \frac{1}{2n^2-n} \sim \sum \left(\frac{1}{n^2}\right) = b_n$$

$$= \text{cgt}$$

$$\lim \frac{a_n}{b_n} = \frac{n^2}{2n^2-n} = \frac{1}{2} \neq 0 \text{ finite}$$

$$\sum \frac{1}{n+\sqrt{n}} \sim \sum \frac{1}{n} \quad b_n$$

$$\lim \frac{a_n}{b_n} = \frac{n}{n+\sqrt{n}} \to 1 \quad \neq 0 \text{ finite}$$

Ratio test:

$$\lim \frac{a_{n+1}}{a_n} = l$$

$l < 1$ cgt
$l > 1$ div
$l = 1$ test fails

Root test

$$\lim (a_n)^{1/n} = l$$

$l < 1$ cgt
$l > 1$ div
$l = 1$ test fails

$$\sum \frac{x^n}{n} \qquad (a_n)^{1/n} = \frac{x}{n^{1/n}}$$

$$(a_n)^{1/n} = \left(\frac{x^n}{n}\right)^{1/n}$$

imp. $\lim p^{1/n} = 1 \leftarrow p = \text{fixed}$

$$n^{1/n} = 1$$

$$\lim (a_n)^{1/n} = \lim \frac{x}{n^{1/n}} = x \quad l$$

$0 < x < 1$ cgt

$x > 1$ div

n ..

$0 < x < 1$ cgt

$x > 1$ div.

$\sum (-1)^n$ $\times$ +ve terms

$= \cancel{1}\ \underbrace{-1 \quad +1}_{0} \ \underbrace{-1 \quad +1}_{0} + \cdots\cdots$

$s_{2n} = 0 \qquad s_{2n+1} = -1$

Imp: If $a_{2n} \to l$ & $a_{2n+1} \to l \Rightarrow a_n \to l$

$\sum (-1)^n \to$ osci. finitely

$\sum (-1)^n n \qquad s_n = \begin{cases} n & \text{even} \to \infty \\ -n & \text{odd} \to -\infty \end{cases}$

osci. infinitely

Alternating series: If $a_n > 0$ then

$\sum (-1)^n a_n$ is alternating series

Leibniz test: If $a_n \downarrow$ ing & $\lim a_n = 0$

then $\sum (-1)^n a_n$ converges.

Eg. (1) $\sum (-1)^n \qquad a_n = 1/n \qquad \lim a_n \to 0$

Eg. (1) $\sum \frac{(-1)^n}{n}$ $\quad a_n = 1/n \quad \lim a_n \to 0$

$\downarrow$ cgt. $= \{ 1, \frac{1}{2}, \frac{1}{3} \dots \}$

(2) $\sum \frac{(-1)^n}{\log n}$ $\quad a_n = \frac{1}{\log n} \quad \lim a_n = 0$

$\downarrow$ cgt.

i.e. $a_{n+1} < a_n$??

$\frac{1}{\log(n+1)} < \frac{1}{\log n}$??

$\log n < \log(n+1)$??

log $\uparrow$ ing $\Rightarrow$ $n < n+1$

(3) $\sum (-1)^n \left(1 + \frac{1}{n}\right)$ $\quad a_n = 1 + \frac{1}{n}$

$a_n = \{ 2, \frac{3}{2}, \frac{4}{3} \dots \}$ $\downarrow$ ing

$\lim a_n = 1 \neq 0$

Eg. 1) $\sum \left| \frac{(-1)^n}{n} \right|$ cgt. $\sum \frac{1}{n}$ div.

$\rightarrow$ cond. cgt

$\sum \frac{(-1)^n}{n^2}$ cgt $\sum \frac{1}{n^2}$ cgt

$\rightarrow$ abs cgt series

Abso. cgt series: If $\sum a_n$ is a series of real no. & $\sum |a_n|$ is cgt $\Rightarrow \sum a_n$ is abs. cgt.

Condi. cgt: If $\sum a_n$ is cgt but $\sum |a_n|$ is div then $\sum a_n$ is cond. cgt.

imp.

Eg. $\frac{1}{1^p} - \frac{1}{2^p} + \frac{1}{3^p} - \cdots\cdots$

$\sum \left| \frac{(-1)^n}{n^p} \right|$ for what values of p abs. cgt / cond. cgt.

$\downarrow$

div. $\rightarrow$ cgt. $\sum \frac{1}{n^p}$

$p \leq 1$ div $\rightarrow$ cond. cgt

$p > 1$ cgt.

$\downarrow$ abs cgt.